

Статистика

Гистограмма статистики MS SQL:

```
USE stat_11;  
GO  
EXEC show_statistics 'AccAg7822', 'AccAg7822_ByPeriod_199' with histogram
```

	RANGE_M_KEY	RANGE_ROWS	EQ_ROWS	DISTINCT_RANGE_ROWS	AVG_RANGE_ROWS
1	4010-10-27 15:22:04.000	0	2	0	1
2	4011-01-05 00:00:00.000	92	202	27	3.407408
3	4011-01-12 00:00:00.000	440	114	28	16
4	4011-01-19 00:00:00.000	328	130	25	13.12
5	4011-01-26 00:00:00.000	307	137	16	19.1875
6	4011-01-27 00:00:00.000	379	179	17	22.29412
7	4011-02-01 00:00:00.000	411	181	21	13.25806
8	4011-02-03 00:00:00.000	239	183	22	9.5
9	4011-02-06 00:00:00.000	341	187	25	13.64
10	4011-02-09 00:00:00.000	52	95	13	4
11	4011-02-15 00:00:00.000	436	216	32	12.6875
12	4011-02-17 00:00:00.000	216	156	21	10.28571
13	4011-02-22 00:00:00.000	394	168	34	11.58823
14	4011-02-24 00:00:00.000	158	164	14	11.28571
15	4011-02-28 00:00:00.000	230	106	19	12.10526
16	4011-03-01 00:00:00.000	192	236	26	7.68
17	4011-03-03 00:00:00.000	179	228	24	7.458333
18	4011-03-07 00:00:00.000	383	192	26	14.73077
19	4011-03-10 00:00:00.000	238	192	12	19

Обновление статистики постгрес

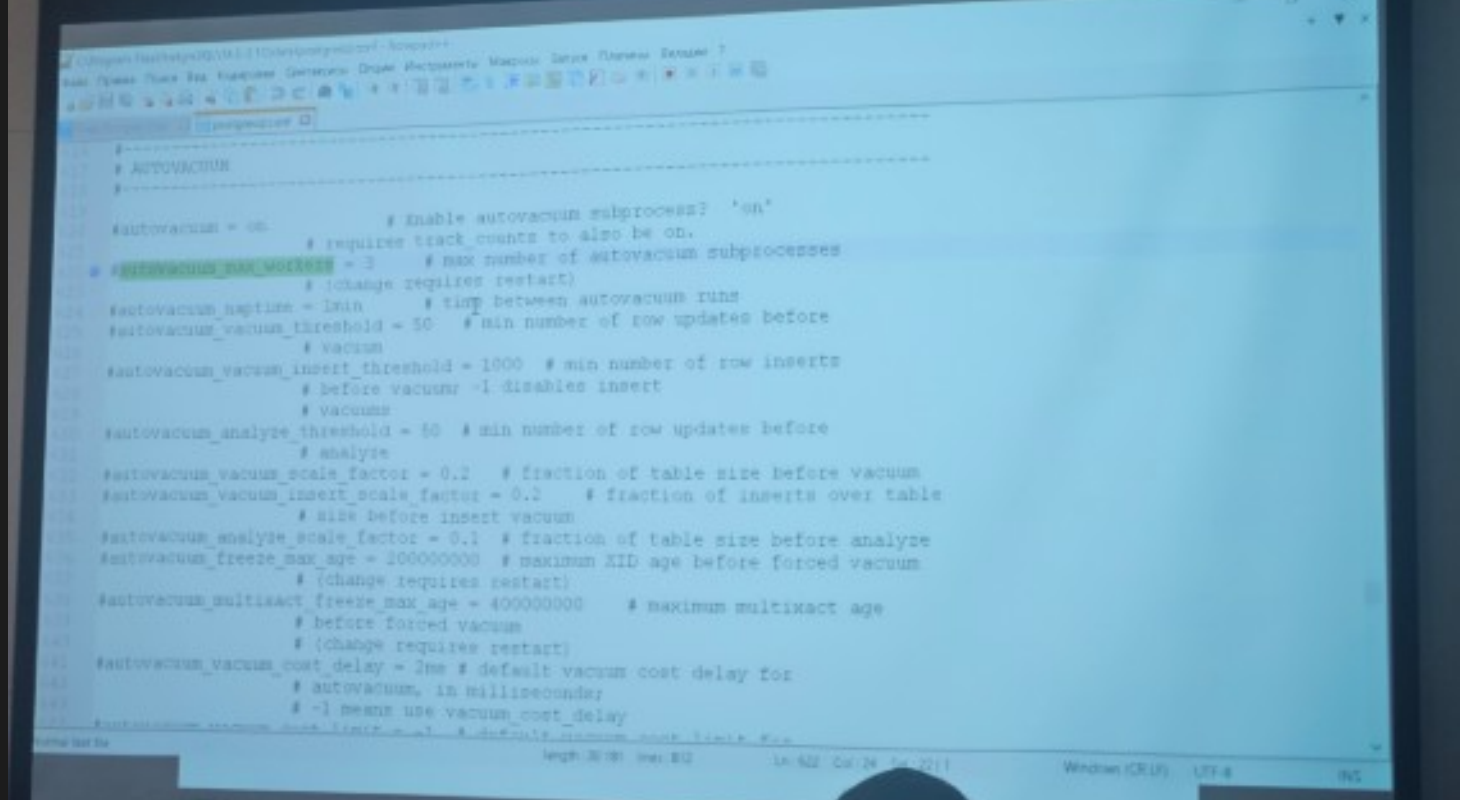
3. Как обновить статистику для Postgres:

TPProBUP@postgres@PostgreSQL Database Server 10.5-9.1C(x64)

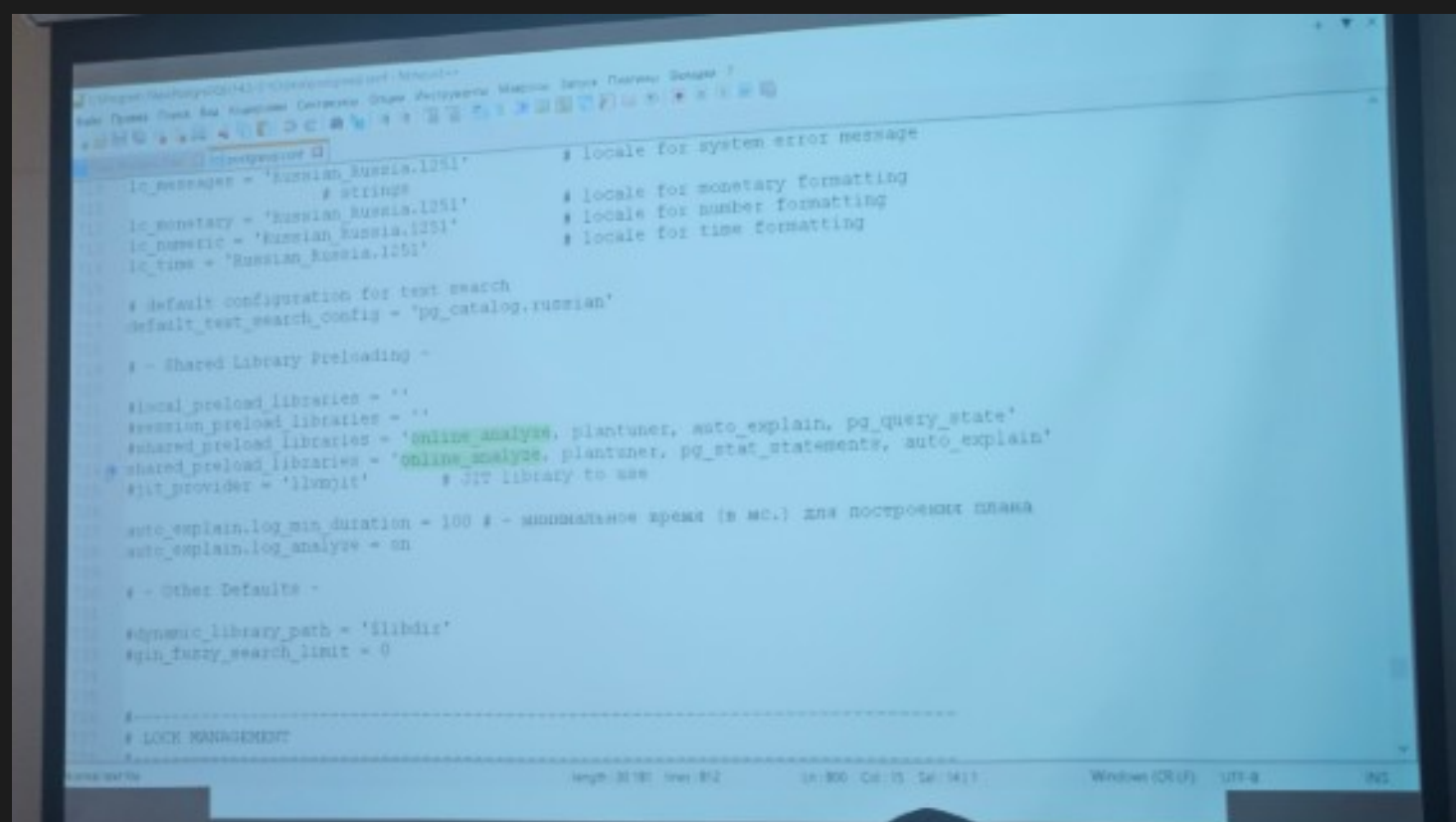
Query Editor Query History

```
1 -- полностью обновить статистику для текущей базы  
2 ANALYZE  
3  
4 -- обновить статистику для таблицы _document18 текущей базы  
5 ANALYZE _document18  
6  
7 -- удалить старые версии строк и обновить статистику для текущей базы  
8 VACUUM ANALYZE  
9  
10 -- удалить старые версии строк, скать и обновить статистику для _document18 текущей базы  
11 -- на время VACUUM таблица будет заблокирована  
12 VACUUM FULL _document18
```

Настройки vacuum



online analyze для временных таблиц. Не использовать с 8.3.13



Обновление статистики MSSQL

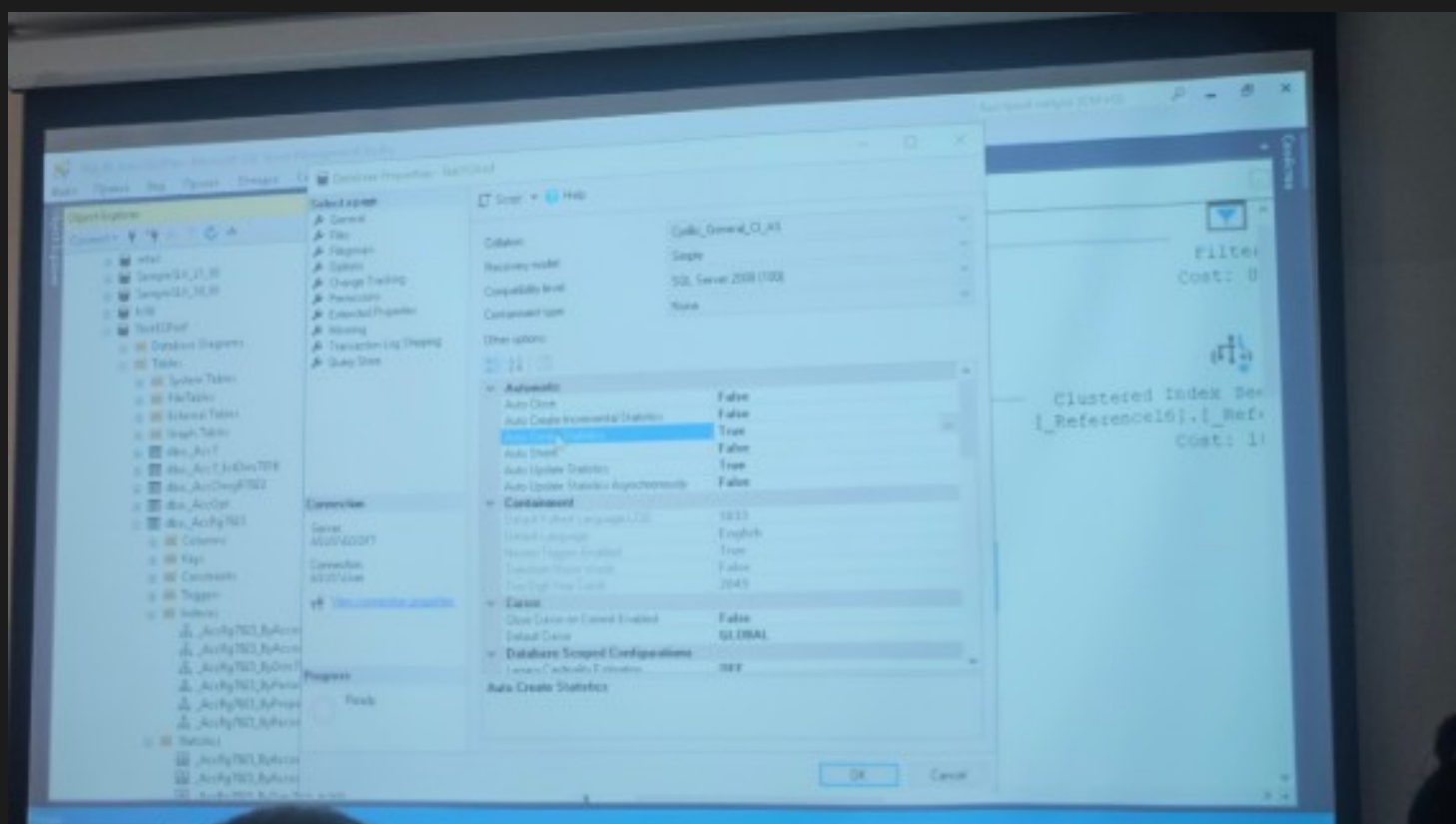
1. Как обновить статистику для MS SQL скриптом:

```

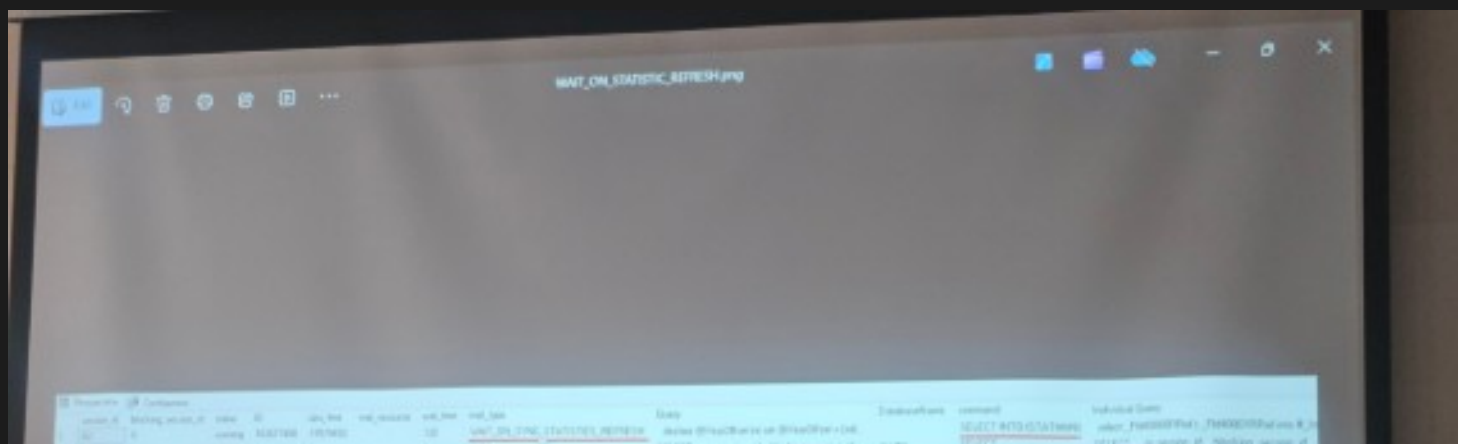
use Test1CProf
go
-- обновить статистику для всех таблиц базы
sp_msforeachtable N'UPDATE STATISTICS ? WITH FULLSCAN'
go
-- обновить статистику для определенной таблицы базы
UPDATE STATISTICS _Acc7 WITH FULLSCAN

```

Авто статистики мssql



Обновляется статистика при выполнении запроса



Как мssql обновляет статистику

<https://learn.microsoft.com/en-us/sql/relational-databases/statistics/statistics?view=sql-server-ver16>

Появление обновлений статистики MSQL.xlsx - Excel

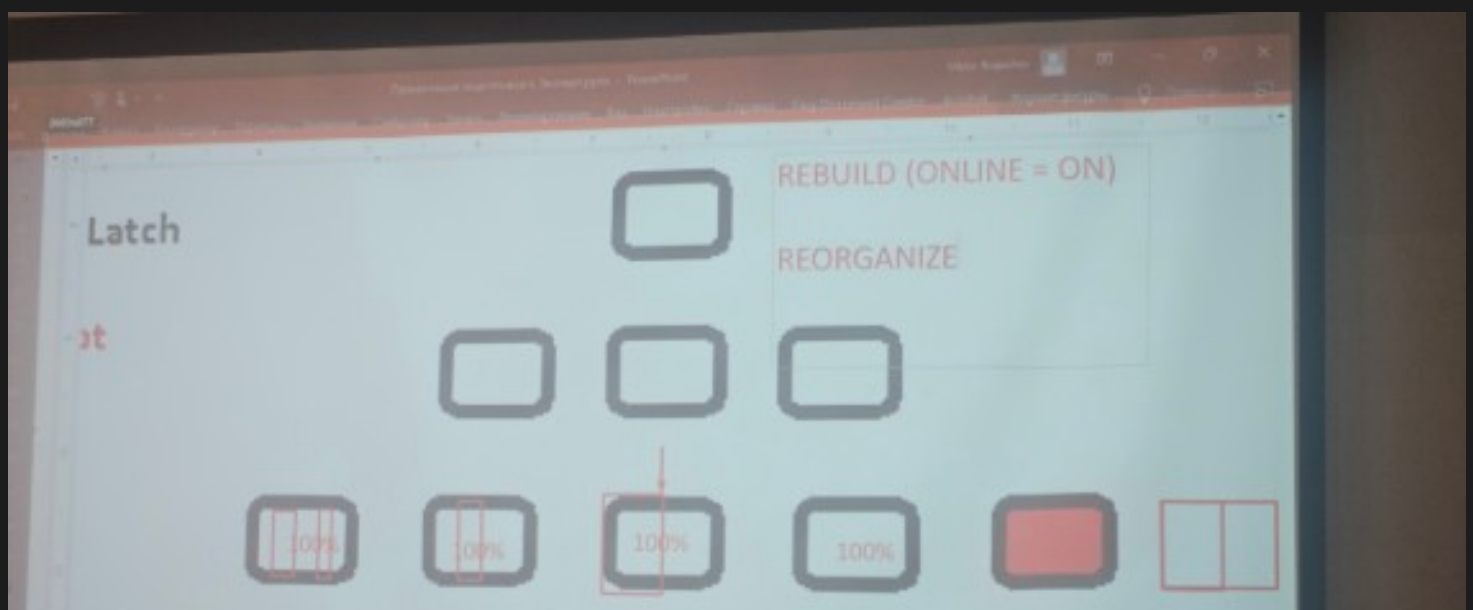
Число изменений в таблице, когда срабатывает автоматическое обновление статистики

<https://learn.microsoft.com/en-us/sql/relational-databases/statistics/statistics?view=sql-server-ver16>

Rows in table	MSQL 2014 and lower	5% updates	2% updates	MSQL 2016 Formula $\text{SQRT}(1000 * \text{RowsInTable})$	% updates for 2016 Formula
1 000	700	50	20	1 000	
10 000	2 500	500	200	3 162	
100 000	20 500	5 000	2 000	10 000	
1 000 000	200 500	50 000	20 000	31 623	
10 000 000	2 000 500	500 000	200 000	100 000	
100 000 000	20 000 500	5 000 000	2 000 000	316 228	
1 000 000 000	200 000 500	50 000 000	20 000 000	1 000 000	
10 000 000 000	2 000 000 500	500 000 000	200 000 000	3 162 278	

Обслуживание базы.

Индексы



Fillfactor - заполненность индексов при перестроении

Постгрес обслуживание:

3. Операции обслуживания баз Postgres

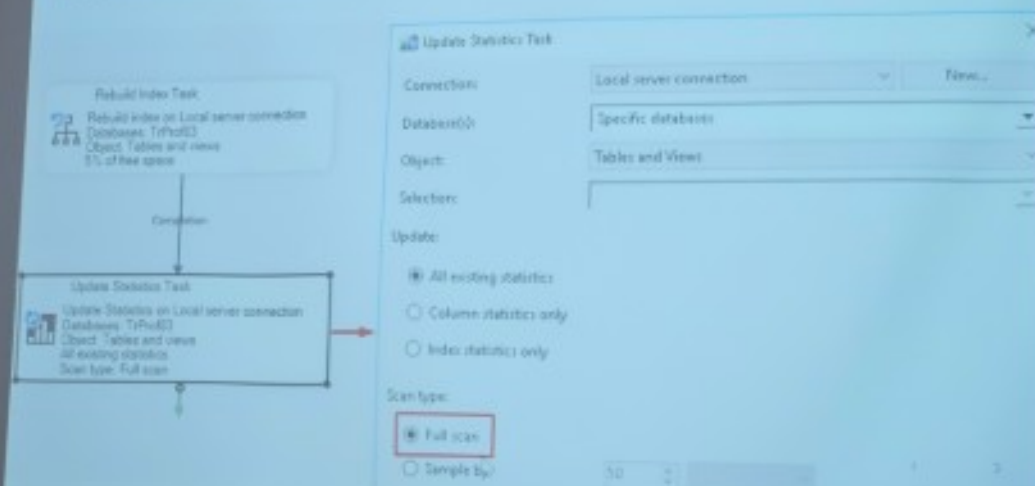
1. `reindex` (блокирует таблицы)
? `reindex concurrently` ?
2. `analyze`
3. `vacuum full` (блокирует таблицы)

Частота выполнения подбирается индивидуально для каждой базы

Обновление статистики мssql

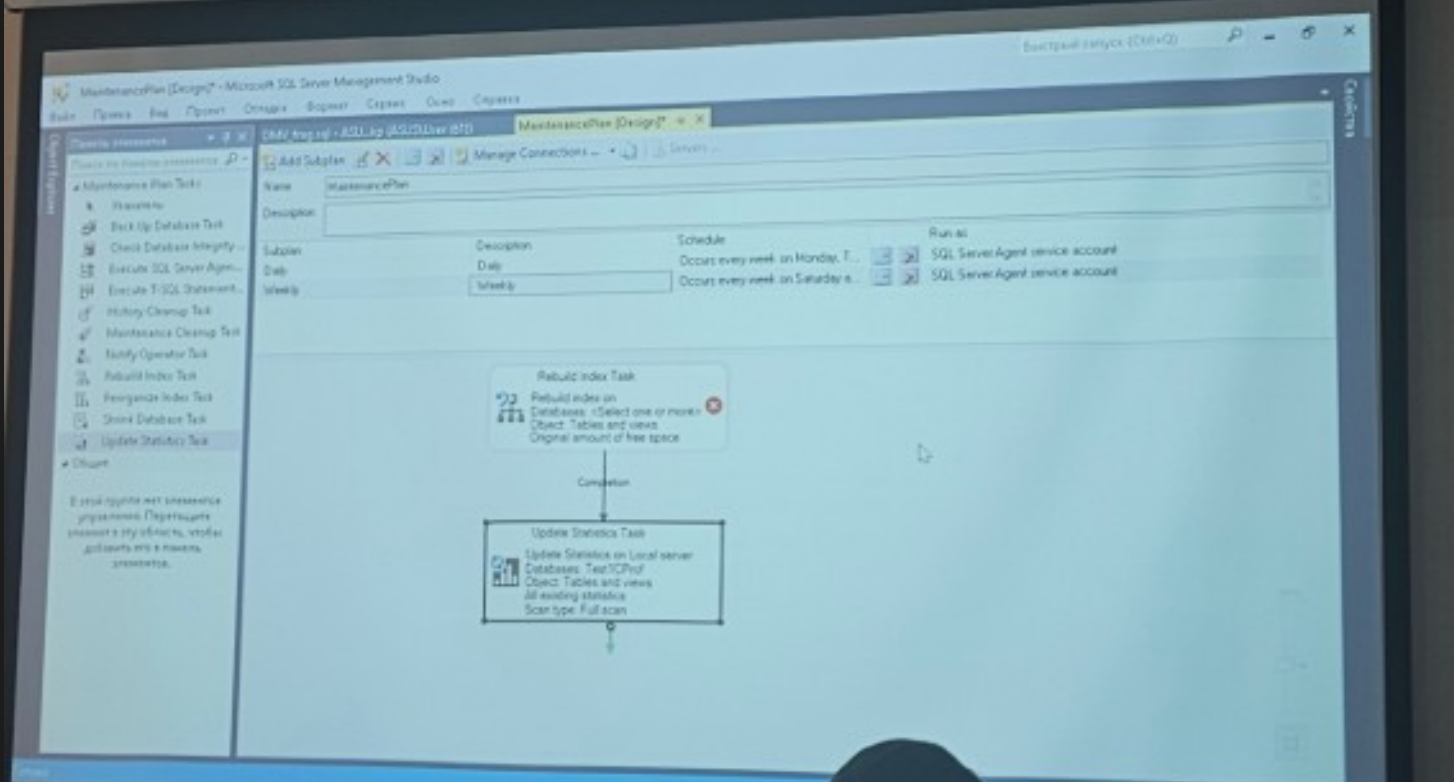
2. Обслуживание индексов и статистики в плане обслуживания MSSQL

Частота выполнения подбирается индивидуально для каждой базы

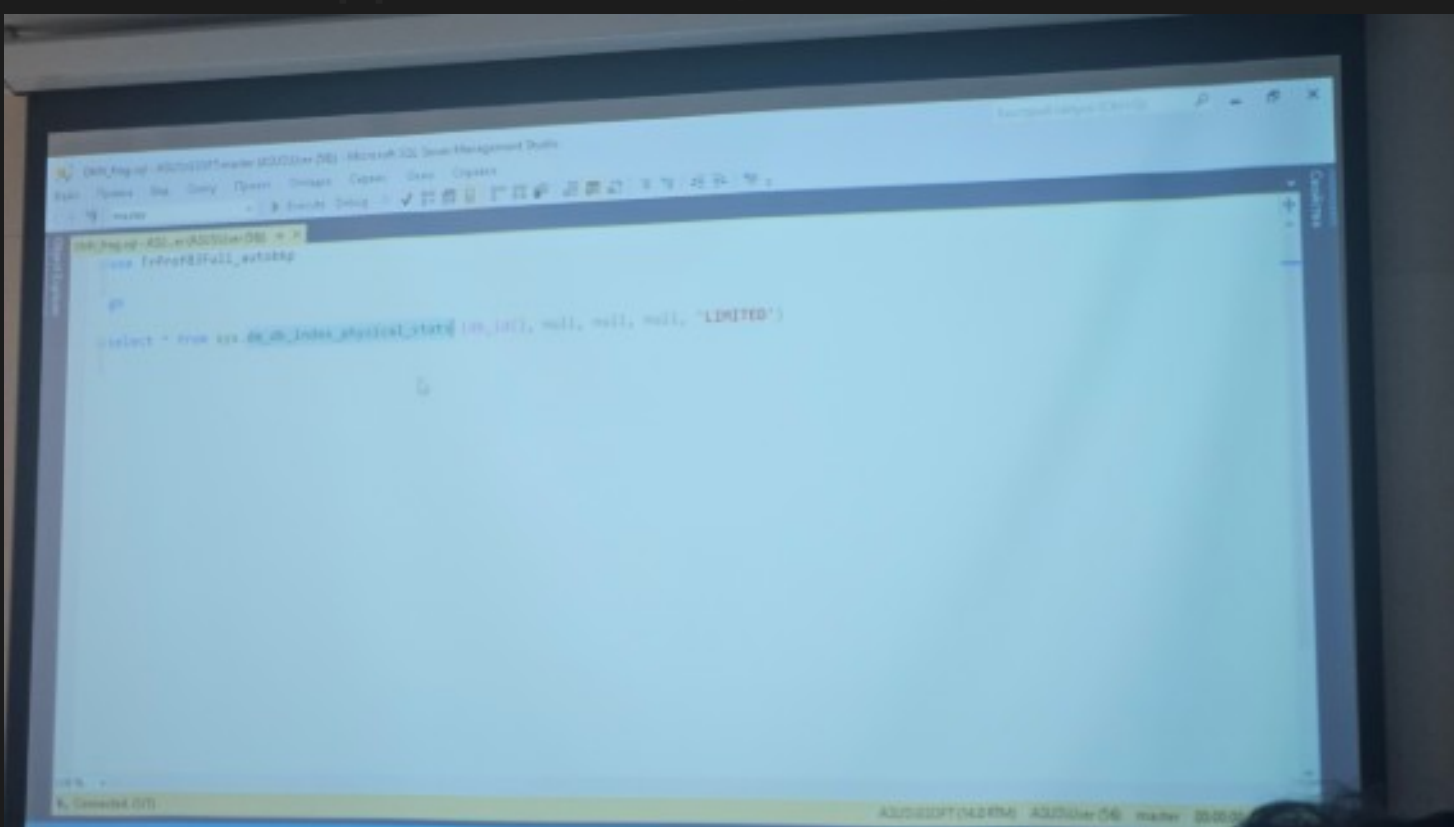


Maintenance plan

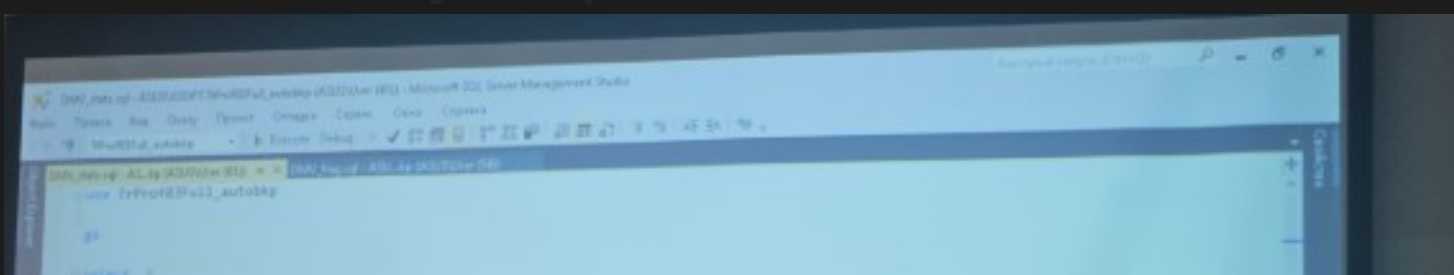
<https://ola-hallengren.com/>



Степень фрагментации индексов



На сколько устарела статистика



```

from sys.objects as obj
inner join sys.stats as stat
on stat.object_id = obj.object_id
cross apply
sql_db_stats_properties(stat.object_id, stat.stats_id) as sp
where obj.type in ('U','V')

```

name	object_id	stats_id	stats_id	parent_obj_id	type	type_desc	create_date	modify_date	is_ms_shipped	is_published	is_schema_published	stats_id
Index1	205702	NUL	1	0	U	USER_TABLE	2019-03-22 12:57:17.727	2021-09-21 18:49:52.667	0	0	0	205702
Index2	205703	NUL	1	0	U	USER_TABLE	2019-03-22 12:57:17.727	2021-09-21 18:49:52.667	0	0	0	205703
Index3	205704	NUL	1	0	U	USER_TABLE	2019-03-22 12:57:17.727	2021-09-21 18:49:52.667	0	0	0	205704
Index4	205705	NUL	1	0	U	USER_TABLE	2019-03-22 12:57:17.727	2021-09-21 18:49:52.667	0	0	0	205705
Index5	205706	NUL	1	0	U	USER_TABLE	2019-03-22 12:57:17.727	2021-09-21 18:49:52.667	0	0	0	205706
Index6	205707	NUL	1	0	U	USER_TABLE	2019-03-22 12:57:17.727	2021-09-21 18:49:52.667	0	0	0	205707
Index7	205708	NUL	1	0	U	USER_TABLE	2019-03-22 12:57:17.727	2021-09-21 18:49:52.667	0	0	0	205708
Index8	205709	NUL	1	0	U	USER_TABLE	2019-03-22 12:57:17.727	2021-09-21 18:49:52.667	0	0	0	205709
Index9	205710	NUL	1	0	U	USER_TABLE	2019-03-22 12:57:17.727	2021-09-21 18:49:52.667	0	0	0	205710
Index10	205711	NUL	1	0	U	USER_TABLE	2019-03-22 12:57:17.727	2021-09-21 18:49:52.667	0	0	0	205711

Query executed successfully.

SQL Server Enterprise Edition (64-bit) - Microsoft SQL Server Management Studio

Server: MSSQLSERVER; Database: master; User: sa; Password: ;

SQL Server Enterprise Edition (64-bit) - Microsoft SQL Server Management Studio

Server: MSSQLSERVER; Database: master; User: sa; Password: ;

Обслуживание многопоточно

```

-- Insert into @index_exception values('UserWorkHistory', 1, 0, 0, 0) -- do not rebuild or reorganize indexes of this table
-- Insert into @index_exception values('Document123', 0, 3, 0, 0) -- service indexes of this table only at wednesdays
-- Insert into @index_exception values('InfoBg20123', 0, 0, 0, 90) -- set fillfactor = 90 for indexes of this table (in case of v
rebuild)

-- Insert into @stat_exception values('InfoBg20123', 1, 0, 0) -- do not update statistics for this table
-- Insert into @stat_exception values('Document123', 0, 0, 20) -- update statistics for this table only at 20 date of each
month

-- Insert into @disabled_indexes values('ConfigChngR_ExtProps', '_ConfigChngR_ExtProps_IntKeyInd', 'enable') -- enable this index, if
it was disabled

declare @RunOfStreams int set @RunOfStreams = 10 -- Number of parallel streams to service indexes and statistics
declare @MaxDOP varchar(max) set @MaxDOP = '0' -- Max degree of parallelism (if possible)
declare @DebugMode int set @DebugMode = 0

declare @Logging_Path varchar(max) set @Logging_Path = 'E:\SQL_DATA' -- Path for logging

declare @Create_jobs_manually int set @Create_jobs_manually = 0 -- Auto or manually create jobs for streams

declare @RowLimit int set @RowLimit = 1 -- Min rows in table to service indexes and statistics

declare @IndexService_Week_Day int set @IndexService_Week_Day = 0 -- If > 0, all indexes will be serviced only this week day
declare @IndexService_Month_Day int set @IndexService_Month_Day = 0 -- If > 0, all indexes will be serviced only this month day
-- If > 0, all indexes CAN be rebuilt only this week day (all other week days indexes could be reorganized)

```

6. Completed 100%

SQL Server Enterprise Edition (64-bit) - Microsoft SQL Server Management Studio

Server: MSSQLSERVER; Database: master; User: sa; Password: ;

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Server: MSSQLSERVER; Database: master; User: sa; Password: ;

```

-- service_week_day - If > 0, statistics of this table can be updated only in this week day
-- service_month_day - If > 0, statistics of this table can be updated only in this month day
declare @stat_exception table (name varchar(max), do_nothing int, service_week_day int, service_month_day int)

-- In this table could be listed indexes to disable, enable (if disabled) or drop (if exists)
-- 'tab_name' - table name
-- 'ind_name' - index name
-- 'ind_action' - may be 'disable', 'drop', 'enable'
declare @disabled_indexes table (tab_name varchar(max), ind_name varchar(max), ind_action varchar(max))

-- Samples of using index or statistics options:

insert into @index_exception values('InfoBg20123', 1, 0, 0, 0) -- service indexes of this table only at wednesdays
insert into @index_exception values('UserWorkHistory', 1, 0, 0, 0) -- do not rebuild or reorganize indexes of this table
insert into @index_exception values('Document123', 0, 3, 0, 0) -- service indexes of this table only at wednesdays
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rebuild)

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-- Insert into @disabled_indexes values('ConfigChngR_ExtProps', '_ConfigChngR_ExtProps_IntKeyInd', 'enable') -- enable this index, if
it was disabled

```

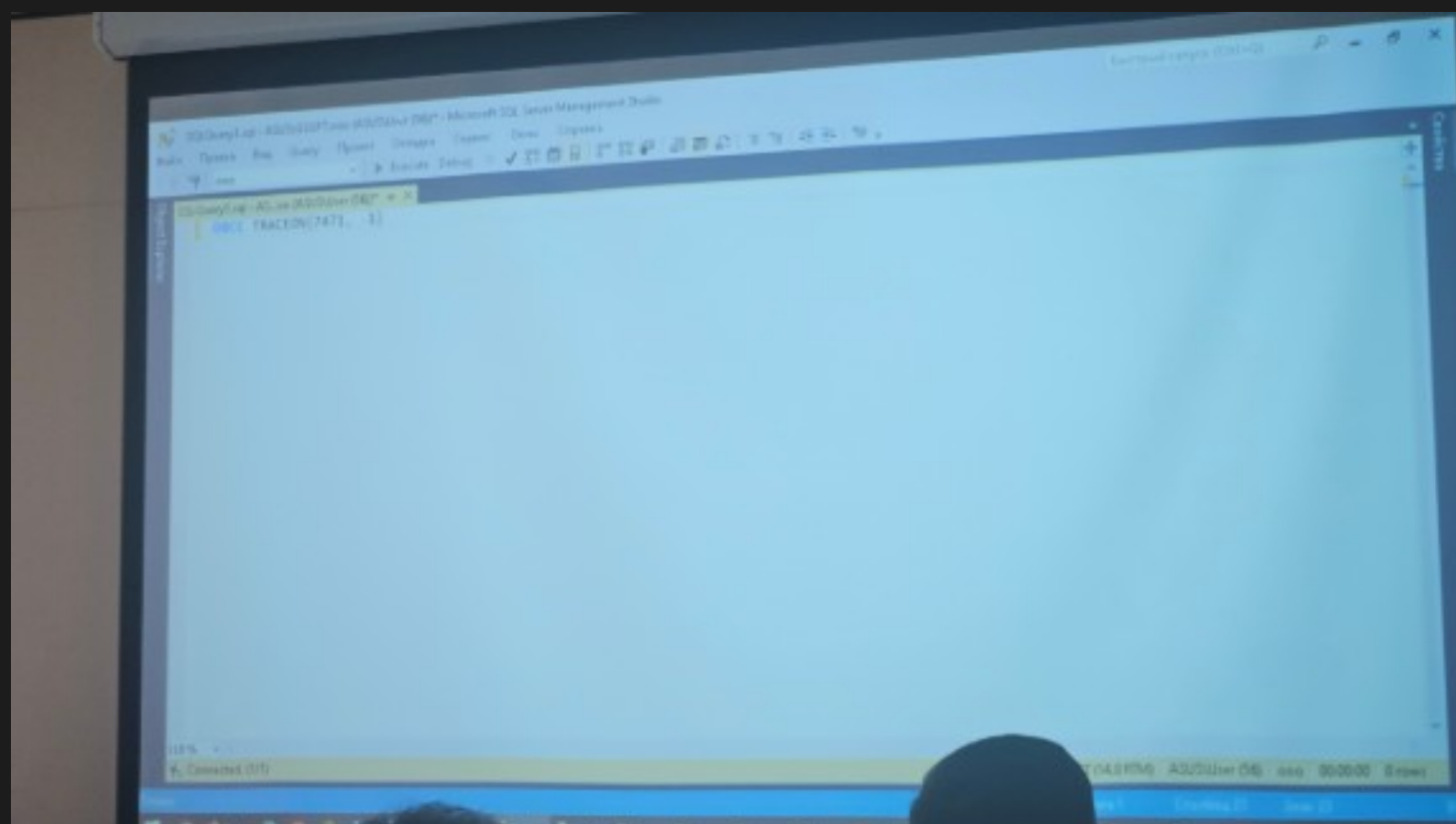
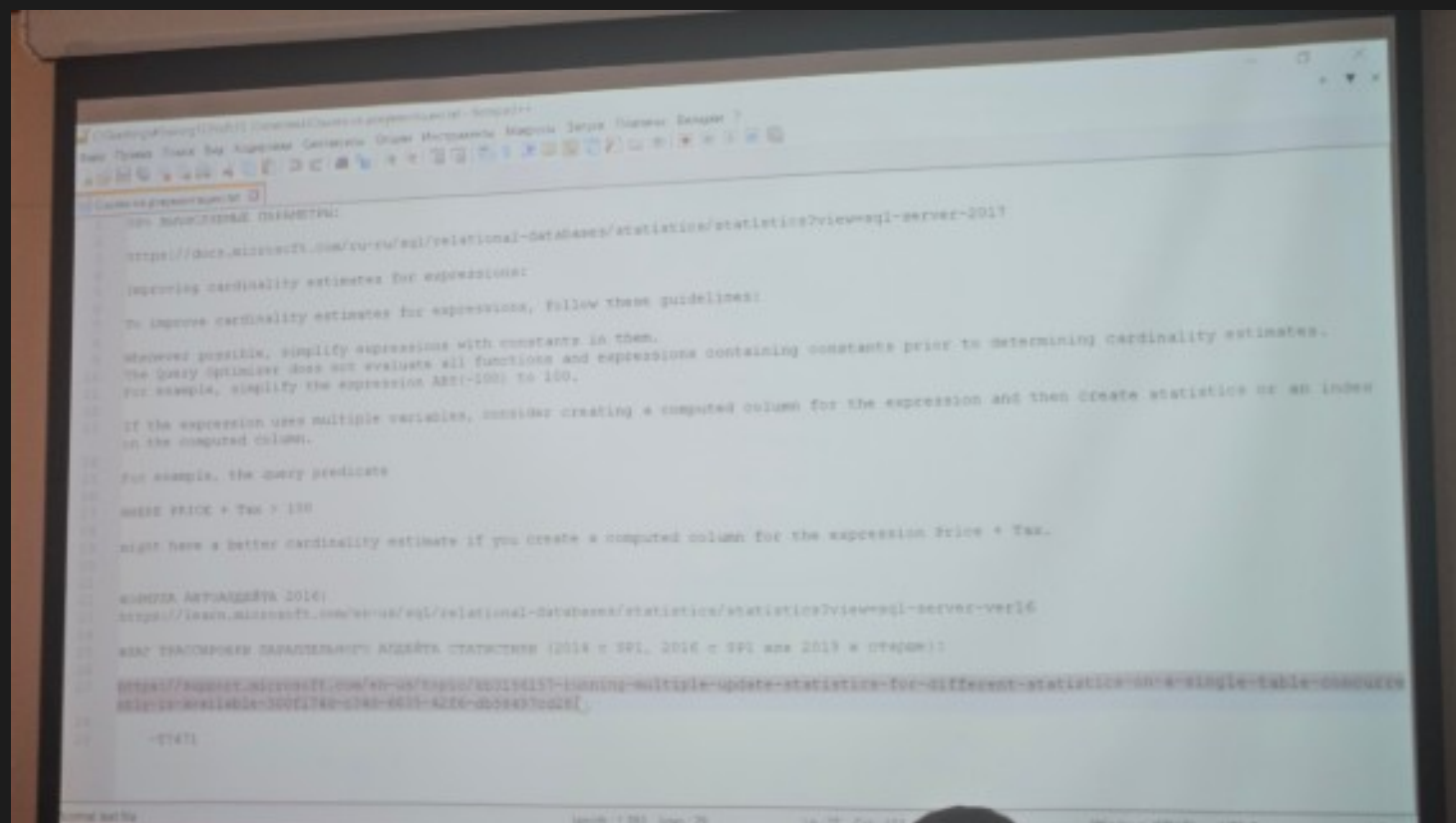
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Server: MSSQLSERVER; Database: master; User: sa; Password: ;

SQL Server Enterprise Edition (64-bit) - Microsoft SQL Server Management Studio

Server: MSSQLSERVER; Database: master; User: sa; Password: ;

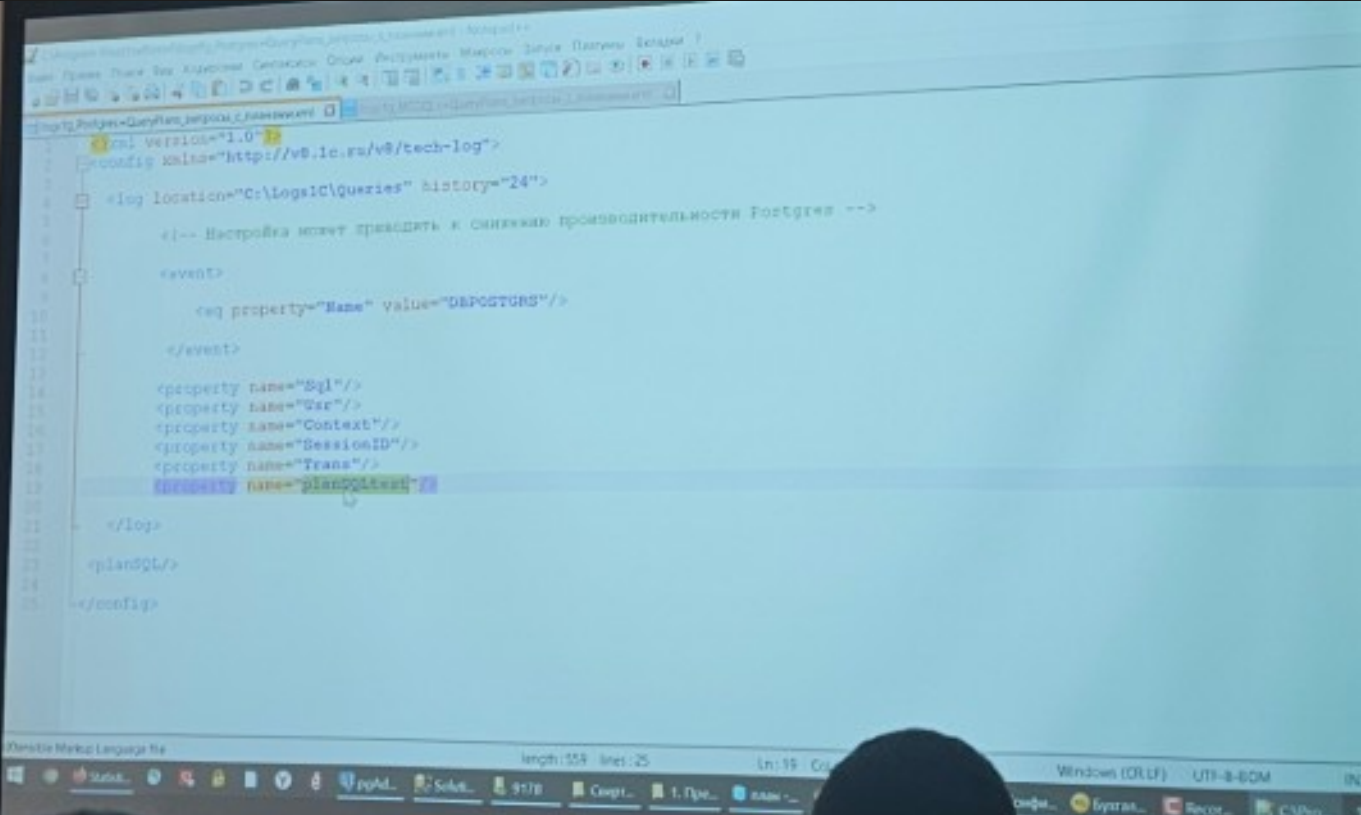
Для больших баз обновление статистики



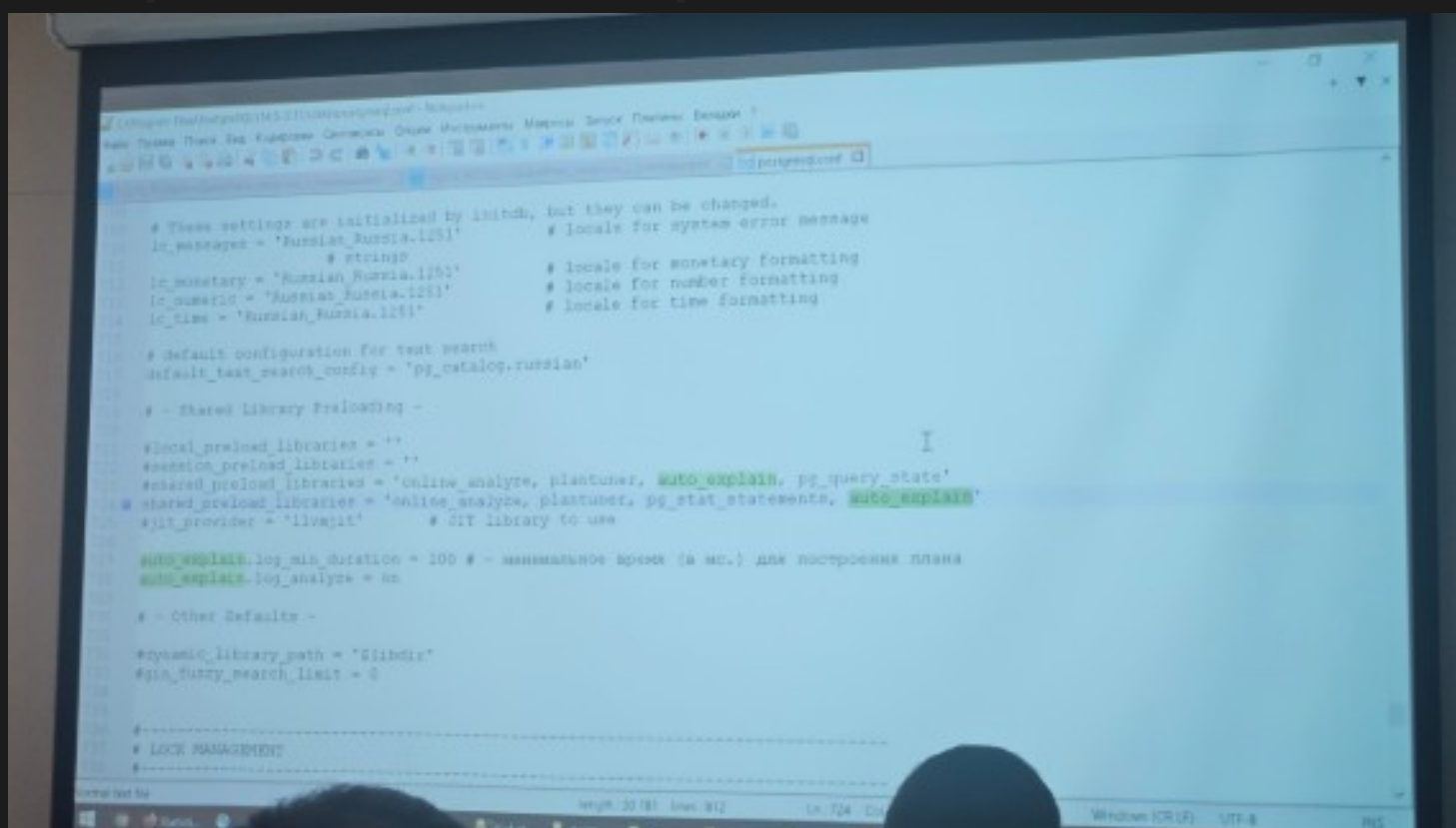
Могут быть блокировки

Планы запросов

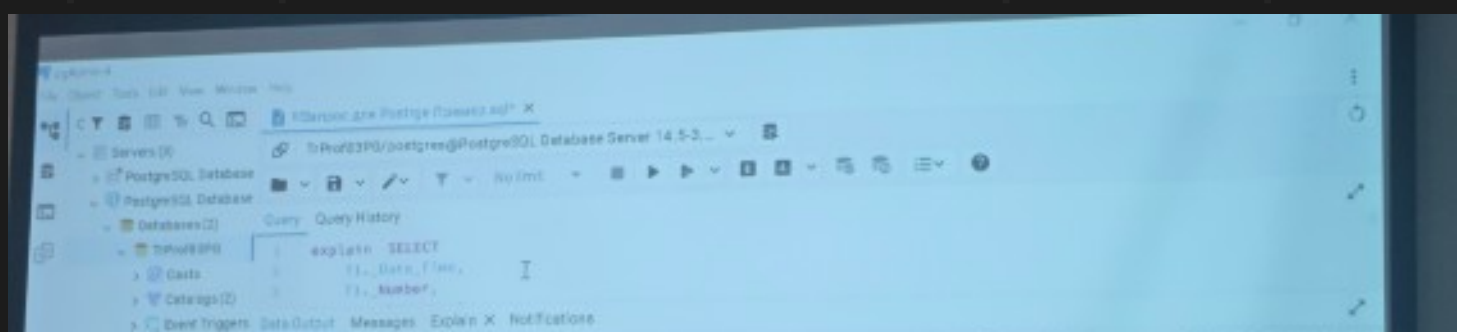
Включение плана запроса в тж

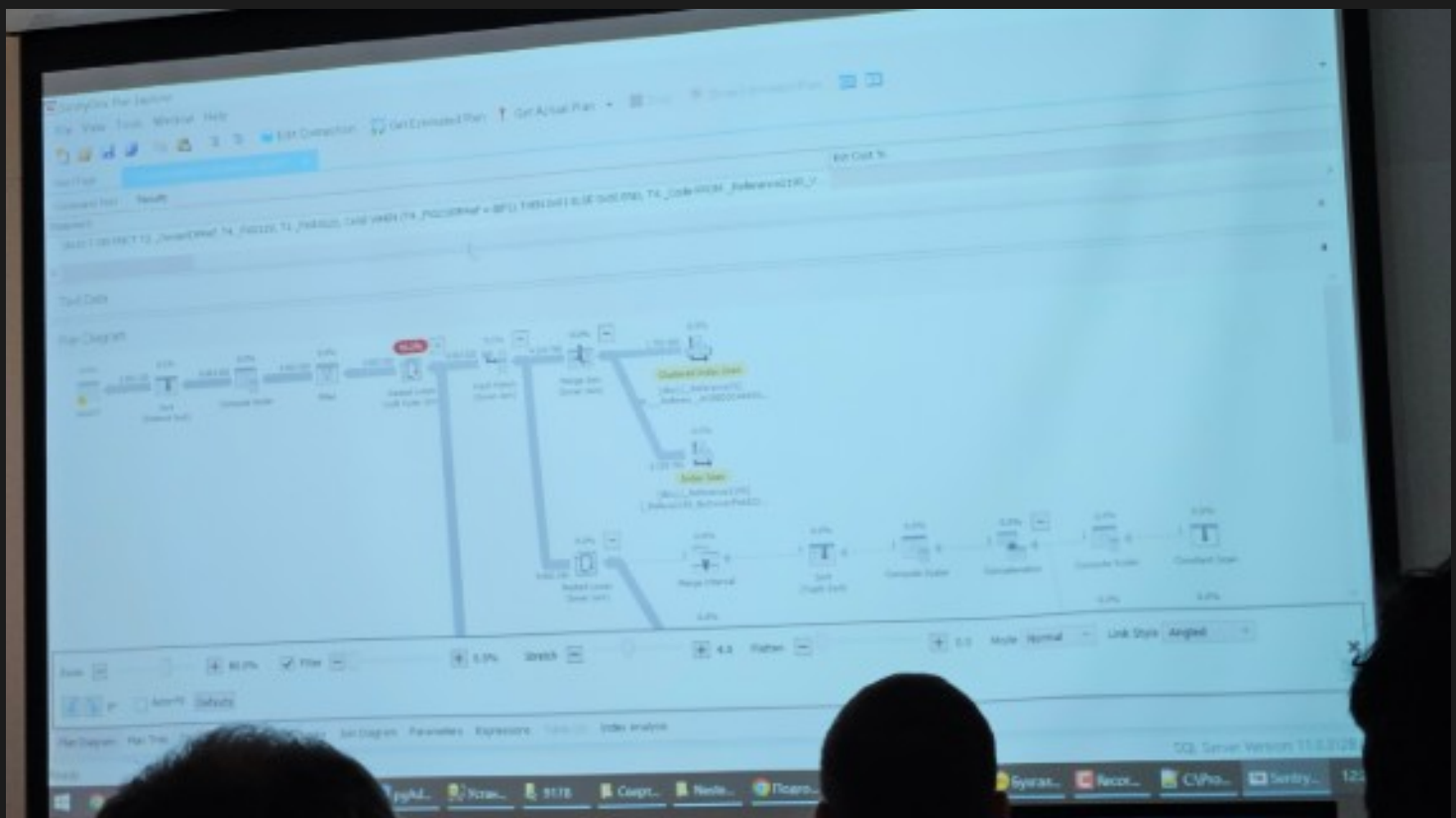


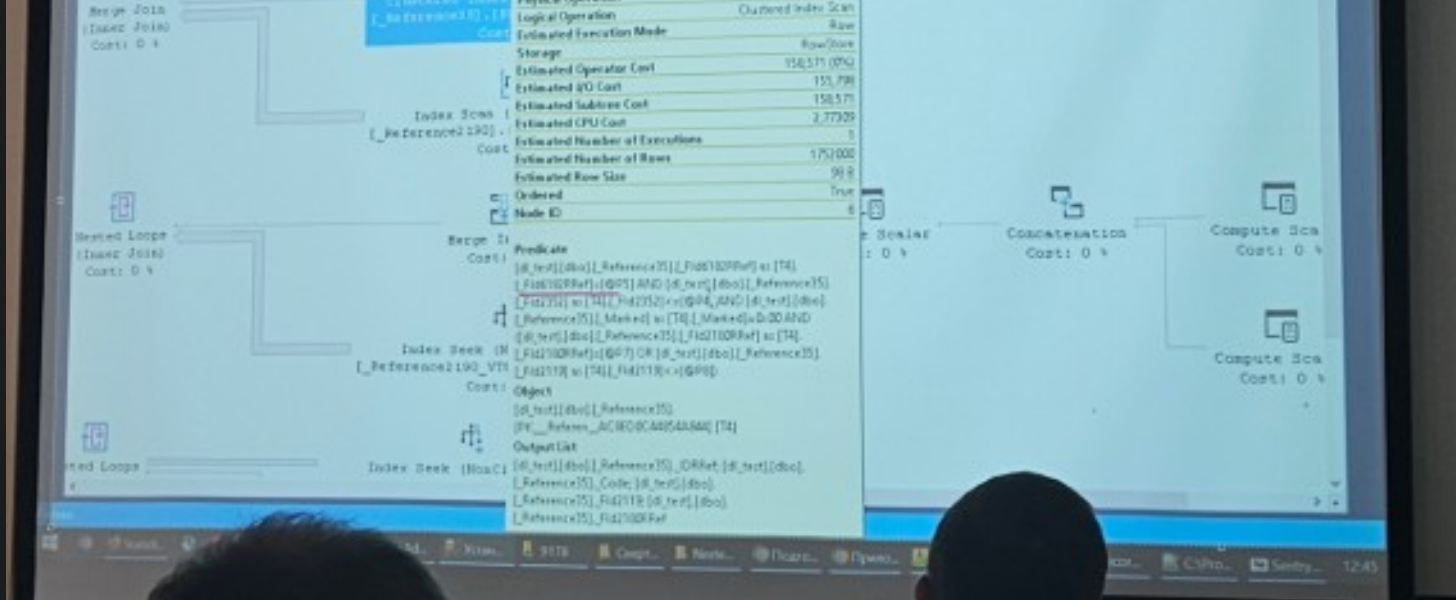
Сбор планов постгри



Предварительный план запроса explain



[illegible]



Настоящий loop - может тормозить когда
вверху толстая стрелка

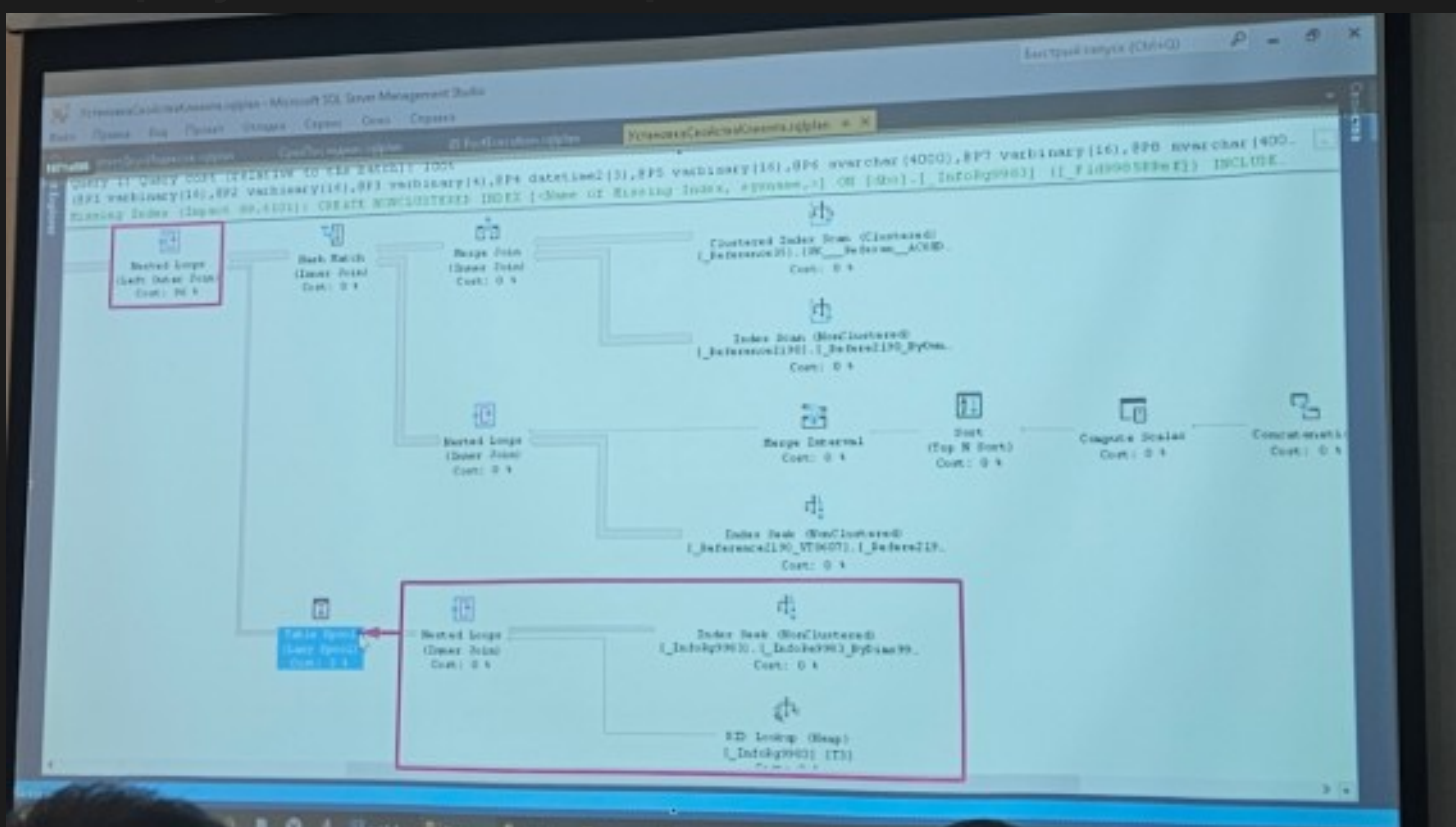


Table spool/materialize - создание
временной таблицы

Рлс - тяжело и плохо

The screenshot shows the SQL Server Enterprise Manager interface. On the left, the 'Object Explorer' pane displays the schema of a table named 'tbl1'. The columns are listed with their data types and constraints. A red box highlights the columns 'tbl1.col1' and 'tbl1.col2'. On the right, the 'Query Results' pane shows the output of a query. The table has columns 'cpu_time', 'physical_reads', 'logical_reads', 'writes', 'timestamp', and 'database'. The data is as follows:

cpu_time	physical_reads	logical_reads	writes	timestamp	database
272	NULL	NULL	NULL	2025-09-18 16:31:42.3952095	
188	NULL	NULL	NULL	2025-09-18 16:31:42.3952095	
187	NULL	NULL	NULL	2025-09-18 16:31:42.4009452	
86	NULL	NULL	NULL	2025-09-18 16:31:42.4074917	

	cpu_time	physical_reads	logical_reads	writes	elapsed_time	database
dfwp_T1	272	NULL	NULL	NULL	2025-09-18 16:31:42.3852395	
dfwp_T1	108	NULL	NULL	NULL	2025-09-18 16:31:42.3859473	
dfwp_T1	167	NULL	NULL	NULL	2025-09-18 16:31:42.4009452	
dfwp_T1	96	NULL	NULL	NULL	2025-09-18 16:31:42.4074917	

